Pyrmit – benchmark report

Feasibility questions on Swedish planning law: tool-using agent vs. classic RAG. Model: GPT-6 Luna via OpenRouter. Runs: 2026-10-07 to 2026-10-17; report generated on 2026-10-17. How the system works and how each benchmark set is built are explained in the **technical report**; this report gives the test conditions and the results.

Summary

- Compared: a **tool-using agent** and a **classic RAG** (one search, one answer), same model and same corpus, on three sets: 14 hand-written case studies, 50 cases from real rulings of the Land and Environment Court of Appeal (MÖD) and 20 questions on a property of a real plan map.
- **The agent wins on everything that requires finding the deciding rule**, and the gap grows with difficulty. Key points: 0.77 vs 0.59 (14 cases), 0.48 vs 0.21 (MÖD), 0.67 vs 0.31 (plankarta). Expected evidence seen: 0.88 / 0.70 / 1.00 vs 0.60 / 0.12 / 0.47.
- **The RAG is about twice as fast and four times cheaper**: 16–24 s per question (first word after 11–18 s), $0.0020 per question, against 31–107 s and $0.0071–$0.0110 for the agent. Its grounding is as high or higher (0.81–0.90): it says less, but what it says is in its sources.
- **Verdict**: all systems score ≥ 0.96 on the 14 cases; on MÖD the agent gets 0.70–0.75 against 0.60 for the RAG, on the plan maps 0.62 against 0.35. The agent's weak spot is still the projects the court refused (39 %–59 % correct): it leans towards “feasible with conditions”.
- **Map reading matters** on the plan-map cases: without the map tools the agent's verdict drops from 0.62 to 0.47 and its key points from 0.67 to 0.54, but the map adds about 54 s per question.
- **Exploration effort is the agent setting that matters**: *low* → *medium* on MÖD raises evidence 0.53 → 0.70, grounding 0.69 → 0.79 and key points 0.43 → 0.48, for +17 s and 2.4× the cost; a *high* final answer brings nothing measurable.

1. Test conditions

1.1 Systems compared

System	What it does	Settings tested
Classic RAG	One rewrite call (English question + Swedish search query, effort low) → one dense search on the Swedish query (arctic-embed-l-v2, municipality filter detected from the question) → the top 10 statute sections and top 10 municipal passages → one answer call, streamed. It cannot open a document, list plans or look at a map.	Answer effort *medium*, streamed (the history table also shows an earlier *high*, non-streamed run)
Agent	Tool loop: exploration turns (search, grep, read a document, statute section, plan registry and plan rules, map crop around a property, PDF page image), tools of a turn run in parallel, then `answer_ready` and one streamed final answer.	Exploration effort *low* / *medium* × answer effort *medium* / *high*; on the plankarta cases, ablations without the map tools

Both answer in the same format (verdict, key rules with [c:id] citations, process, uncertainties) and search the same corpus: the full disk store (15 statutes, the crawled websites of the municipalities, 8,258 indexed detailed plans).

1.2 Benchmark sets

Set	Origin	Cases run	What it tests	Reference
14 case studies	Hand-written from real documents of 5 municipalities (detailed plans of 3 real projects, municipal guidance pages, statutes) and questions from user feedback	14 (all)	Combining national and local rules on common projects	Accepted verdicts, 3–5 key points, expected evidence (statute section or exact municipal excerpt)
MÖD rulings	Real appeals decided by the Land and Environment Court of Appeal (746 rulings since 2011, 390 on our topics); the facts become the applicant's question, asked before the decision	50 of 260 (stratified by outcome, fixed seed)	Legal judgement on real disputes; verdict against the real outcome	Outcome → accepted verdicts; court's reasons → key points; statutes applied → evidence (current wording)
Plankarta	Real plan maps (in force or adopted) of 18 municipalities, a property number printed on the map; the project is designed so that its feasibility depends on the zone rules	20 (high-confidence references)	Reading the map: which zone the property is in, which codes apply	Codes and values read on the map crop + legend by a vision model; evidence = the plan map document

1.3 Protocol and scoring

- Same model for the systems and the judges (GPT-6 Luna via OpenRouter); judge effort medium.
- Each configuration runs 2 passes or more (n = answers = cases × passes); cases run 3 in parallel, so latencies include some contention. Answers are streamed: "1st token" is the time until the first word of the answer.
- Results are written per answer (`data/agentic/bench-*.json`) with the full answer, the tool calls and the judge's comment.

Measure	Definition	How
Verdict	The answer's verdict (feasible / with conditions / not feasible / unclear) is among the accepted verdicts	LLM judge
Key points	Each expected point 0 (missing or contradicted), 1 (partial), 2 (clear); scaled to 0–1	LLM judge
Evidence seen	Share of the expected evidence the system *actually had in front of it* (retrieved passages for the RAG, tool outputs for the agent)	Automatic, no LLM
Evidence cited	Share of the expected evidence the answer *cites*: did the system recognise the decisive passage among what it read? The gap with "evidence seen" is the passages read but not used	Automatic, no LLM
Passages seen	Distinct passages the system had in front of it per question: the reading volume. A precision ("relevant / read") is not measurable: the labels list the decisive passages, not every useful one	Automatic
Grounding	Share of the answer's factual claims supported by the text of the sources it cites	Second LLM judge, full source text
Latency, 1st token, cost	Total time, time to the first answer word, OpenRouter cost of the system's calls (judges excluded)	Measured

> **Limits.** The judge is the system's own model: cheap but possibly lenient, so its scores *compare* settings rather than give absolute quality. MÖD key points and plankarta references are written by an LLM (from the ruling, from the map); no human review yet. A difference below ~0.05 between configurations is within the pass-to-pass noise (section 2.6). **Scope**: every case is a feasibility check of a defined project at a known place; the results say nothing about helping shape a project from a vague idea, multi-turn guidance or finding where a project is possible (technical report, section 7.8).

2. Results

2.1 Overview: classic RAG vs. agent (default setting, medium / medium)

Case set	System	n	Verdict	Key points	Evidence seen	Evidence cited	Grounding	Passages seen	Latency	1st token	Cost / question
14 cases	RAG	28	1.00	0.59	0.60	0.57	0.90	20	16 s	11 s	$0.0017
	Agent (medium / medium)	28	1.00	0.77	0.88	0.69	0.89	69	31 s	24 s	$0.0071
MÖD	RAG	100	0.60	0.21	0.12	0.12	0.88	20	23 s	17 s	$0.0020
	Agent (medium / medium)	200	0.70	0.48	0.70	0.53	0.79	78	43 s	35 s	$0.0086
plankarta	RAG	40	0.35	0.31	0.47	0.28	0.81	20	24 s	18 s	$0.0020
	Agent (medium / medium)	40	0.62	0.67	1.00	0.90	0.68	66	107 s	99 s	$0.0110

The agent reads 38–80 passages per question through 12–27 tool calls, each with its own query; the RAG reads the 20 passages of a single search. The number is not the point (a case has 1 to 2 expected passages): the RAG's single query, written from the user's facts, rarely lands on the statute section or the plan map that decides the case, whereas the agent first works out which rule applies, then looks it up. Of what it sees, the agent cites about three quarters of the expected evidence (evidence cited vs seen); the RAG cites most of what it finds, but outside the 14 case studies it finds little.

2.2 14 case studies

History (5-municipality sub-corpus, successive versions):

Version	Verdict	Key points	Evidence	Grounding	Latency	1st token	Cost
Classic RAG (1 search, high-effort answer, not streamed)	1.00	0.56	0.58	0.84	30 s	–	$0.0021
Agent v1 (high effort throughout)	1.00	0.75	0.86	0.68	64 s	–	$0.0057
Agent v2 (citeable sources, medium effort)	1.00	0.77	0.96	0.86	39 s	39 s	$0.0042
Agent v3 (low exploration, medium answer)	1.00	0.68	0.73	0.82	19 s	13 s	$0.0025
Agent v4 (v3 + read the key municipal document)	1.00	0.65	0.84	0.84	23 s	17 s	$0.0036

v1 → v2: citeable identifiers in every tool output and a “no claim without a source read” rule (grounding 0.68 → 0.86). v2 → v3/v4: parallel tools, `answer_ready`, streamed answer and *low* exploration effort: latency halved, lower quality. The RAG of this table ran with a *high*, non-streamed answer; it is re-measured below at *medium*, streamed.

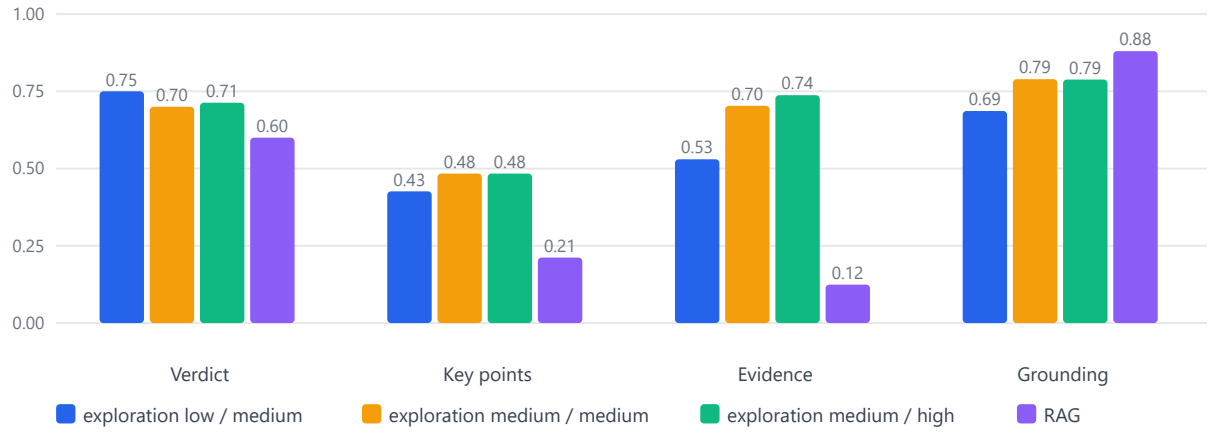
Effort matrix (full corpus on disk):

Configuration	n	Verdict	Key points	Evidence seen	Evidence cited	Grounding	Passages seen	Tool calls	Latency	1st token	Cost / question
Agent, exploration low / answer medium	28	0.96	0.70	0.84	0.70	0.85	38	12.0	28 s	22 s	$0.0036
Agent, exploration medium / answer medium	28	**1.00**	**0.77**	**0.88**	0.69	0.89	69	20.2	31 s	24 s	$0.0071
Agent, exploration medium / answer high	56	0.98	0.76	0.87	**0.73**	0.85	63	20.2	38 s	31 s	$0.0069
Classic RAG (medium answer)	28	**1.00**	0.59	0.60	0.57	**0.90**	20	0.0	16 s	11 s	$0.0017

2.3 MÖD cases (50 sampled rulings)

Configuration	n	Verdict	Key points	Evidence seen	Evidence cited	Grounding	Passages seen	Tool calls	Latency	1st token	Cost / question
Agent, exploration low / answer medium	100	**0.75**	0.43	0.53	0.39	0.69	40	12.3	26 s	19 s	$0.0036
Agent, exploration medium / answer medium	200	0.70	0.48	0.70	0.53	0.79	78	26.4	43 s	35 s	$0.0086
Agent, exploration medium / answer high	150	0.71	**0.48**	**0.74**	**0.54**	0.79	80	26.7	51 s	42 s	$0.0092
Classic RAG (medium answer)	100	0.60	0.21	0.12	0.12	**0.88**	20	0.0	23 s	17 s	$0.0020

Best value of each column in green. n = number of answers (cases × passes).



Case by case on the 50 MÖD cases, `medium` exploration beats `low` on 25 cases, does worse on 3, and ties (±0.05) on 22.

Verdict by actual outcome (in brackets: number of answers; “open outcome” = remitted or fact-dependent decision, accepted verdicts “with conditions” / “unclear”):

Configuration	Project refused (correct verdict)	Project granted	Open outcome	On refusals, answers “with conditions”
Agent, exploration low / answer medium	0.59 (32)	0.74 (34)	0.91 (34)	0.28
Agent, exploration medium / answer medium	0.39 (64)	0.79 (68)	0.90 (68)	0.44
Agent, exploration medium / answer high	0.50 (48)	0.73 (51)	0.90 (51)	0.38
Classic RAG (medium answer)	0.53 (32)	0.32 (34)	0.94 (34)	0.03

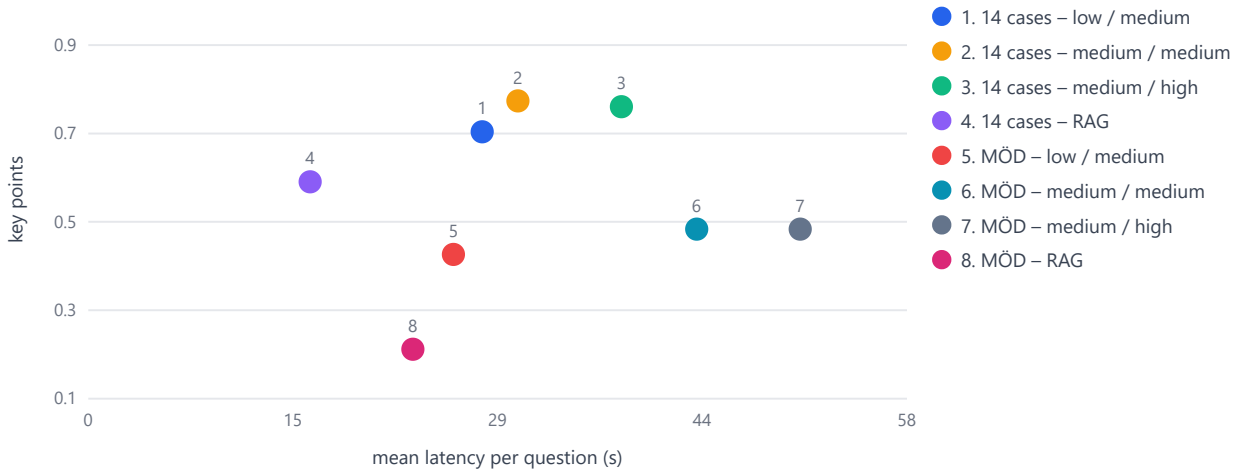
2.4 Plankarta cases (map reading)

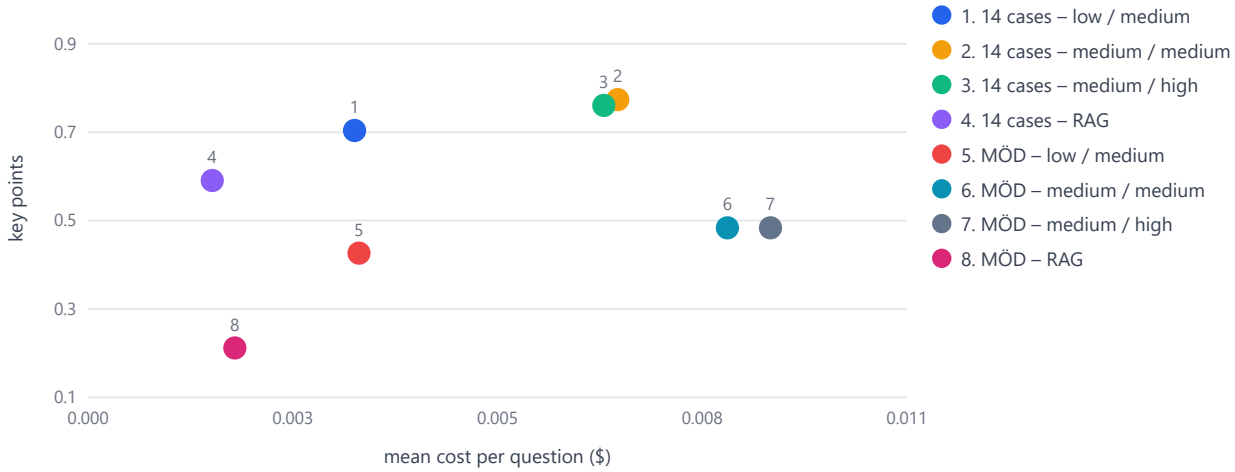
Configuration	n	Verdict	Key points	Evidence seen	Evidence cited	Grounding	Passages seen	Tool calls	Map views	Latency	1st token	Cost / question
Agent, exploration medium / answer medium	40	**0.62**	**0.67**	**1.00**	**0.90**	0.68	66	25.3	2.3	107 s	99 s	$0.0110
Agent, exploration medium / answer medium – no view_map_area	40	0.60	0.57	0.97	0.82	0.82	77	26.6	1.7	77 s	70 s	$0.0102
Agent, exploration medium / answer medium – no map reading at all	40	0.47	0.54	**1.00**	0.82	**0.84**	75	24.1	0.0	53 s	45 s	$0.0082
Classic RAG (medium answer)	40	0.35	0.31	0.47	0.28	0.81	20	0.0	0.0	24 s	18 s	$0.0020

Map views = images of the plan map the agent looked at per question (`view_map_area`, `view_pdf_page`). Evidence = the plan map of the case was among the passages seen.

With both map tools the agent gets the verdict right in 0.62 of the answers and covers 0.67 of the key points; without `view_map_area` (whole-page images only) 0.60 / 0.57; with no map at all 0.47 / 0.54: it has to guess the zone from the text. The map views make this set slow (107 s per question). Grounding is lower with the map (0.68 vs 0.84), probably because what is read on the image (which zone the property is in) cannot be checked by the grounding judge, which only gets text. The classic RAG, with one search and no map, has the plan map of the case among its passages in 0.47 of the answers and gets the verdict right in 0.35.

2.5 Quality / speed / cost trade-off (14 cases and MÖD)





2.6 Variance between passes

- Agent, exploration low / answer medium (14 cases): key points per pass 0.74 / 0.67
- Agent, exploration medium / answer medium (14 cases): key points per pass 0.75 / 0.79
- Agent, exploration medium / answer high (14 cases): key points per pass 0.77 / 0.78 / 0.71 / 0.78
- Classic RAG (medium answer) (14 cases): key points per pass 0.58 / 0.60
- Agent, exploration low / answer medium (MÖD): key points per pass 0.43 / 0.43
- Agent, exploration medium / answer medium (MÖD): key points per pass 0.50 / 0.50 / 0.46 / 0.47
- Agent, exploration medium / answer high (MÖD): key points per pass 0.49 / 0.47 / 0.49
- Classic RAG (medium answer) (MÖD): key points per pass 0.21 / 0.21
- Agent, exploration medium / answer medium (plankarta): key points per pass 0.68 / 0.66
- Agent, exploration medium / answer medium – no view_map_area (plankarta): key points per pass 0.56 / 0.58
- Agent, exploration medium / answer medium – no map reading at all (plankarta): key points per pass 0.51 / 0.57
- Classic RAG (medium answer) (plankarta): key points per pass 0.37 / 0.25

Typical gap between two identical passes: a few hundredths; a difference below ~0.05 between configurations is not significant.

3. Reading the results

- **The search strategy separates the systems, not the volume read.** The RAG has 20 slots for 1–2 expected passages, but one query. On the MÖD cases every expected passage is a statute section, among several

thousand, and the question describes facts (a house by a lake, a barn on farmland) rather than the rule: the rewrite rarely reaches the deciding section (evidence 0.12). The agent reasons first ("this is shoreline protection: exemption and its special reasons") and then fetches the section.

- **Where the RAG is enough.** On the 14 case studies (short questions, well-covered municipalities, answers in one municipal page or a well-known section) it reaches the verdict every time, with the best grounding and half the latency; it still covers 0.18 fewer key points than the agent.
- **Verdict calibration.** The RAG is right more often than the agent on refused projects (53 % vs 39 %) but much less on granted ones (32 % vs 79 %): missing the conditions and exceptions that make a project possible, it is more negative rather than better calibrated. The agent's "with conditions" on refusals seems to come from treating the conditions it finds as requirements to meet rather than as blockers.
- **Exploration effort**: at *low* the agent makes ~12 tool calls per question, at *medium* ~20–26, better targeted; it then sees and cites much more of the expected evidence. More exploration also finds more conditions, hence a slightly lower verdict score on MÖD (0.75 → 0.70) despite better evidence.
- **Grounding is not quality on its own.** The RAG has the highest grounding and the lowest key-point coverage: the measure says that stated claims are supported, not that the important ones are there. Read it together with key points and evidence.
- **Difficulty.** The MÖD and plan-map cases separate the systems; the 14 case studies saturate on the verdict and only show differences in key points and evidence.

4. Recommendations

- **Default app setting**: the agent, *medium* exploration and *medium* answer; *low* exploration as a fast mode (≈ 26 s). A *high* answer is not worth its cost.
- **The classic RAG is not a substitute** for feasibility questions beyond the simplest ones; it could serve where speed matters more than completeness (a first quick answer while the agent explores), to be evaluated as such.
- **Verdict prompt**: state that an unmet substantive condition (productive farmland without an essential public interest, shoreline protection without a special reason, a project that requires a detailed plan) makes the project "not feasible as described", and measure it on the refused MÖD cases.
- **Map reading**: keep `view_map_area`; make it cheaper (one crop instead of crop + page, smaller images) since it doubles the time on plan questions.
- **Benchmark**: run the MÖD sample (50 cases, 2 passes) and the plan-map set for every prompt or model change; add sets for the use cases of the product vision (shaping a project from a vague idea, multi-turn guidance), which these sets do not measure.
- **Judge**: have a domain expert review a sample of scores and of plan-map references.

Appendix: reproduce

`python -m agentic.bench --systems agent --store disk --explore-effort medium --effort medium --cases-file cases_caselaw.jsonl --sample 50 --repeats 2 --workers 3` (MÖD; `--systems rag` for the classic RAG; `--cases-file cases_plans.jsonl` for the plankarta cases, `--disable-tools view_map_area` for the ablations) · `python -m agentic.grounding <results>` · `python -m agentic.report`. Total cost of the answers in this report: ~$5.64 (judges and case generation excluded). Result files used: bench-20261007-211220-agent-medium-explore-low.json, bench-20261007-214122-agent-medium-explore-low-cases_caselaw.json, bench-20261007-215229-agent-medium-explore-medium.json, bench-20261007-221849-agent-medium-explore-medium-cases_caselaw.json, bench-20261007-222545-agent-medium-explore-medium-cases_caselaw.json, bench-20261007-223027-agent-high-explore-medium.json, bench-20261007-225332-agent-high-explore-medium.json, bench-20261007-224026-agent-high-explore-medium-cases_caselaw.json, bench-20261007-230229-agent-high-explore-medium-cases_caselaw.json, bench-20261008-134746-agent-medium-explore-medium-cases_plans-no-view_map_area-view_pdf_page.json, bench-20261008-141019-agent-medium-explore-medium-cases_plans-no-view_map_area.json, bench-20261008-143710-agent-medium-explore-medium-cases_plans.json, bench-20261017-133642-rag-medium-explore-low.json,

bench-20261017-134133-rag-medium-explore-low-cases_caselaw.json, bench-20261017-140205-rag-medium-explore-low-cases_plans.json.